Problem Chosen	2025 MCM/ICM Summary Sheet	Team Control Number
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Staircase Sentinel: Unraveling Staircase History and Usage Patterns

Summary

This paper presents an integrated model to analyze the wear patterns of staircases and determine their usage history. Our approach combines quantitative formula derivation, simulation-based visual analysis, and non-destructive measurement techniques. It provides guidance for archaeologists studying worn staircases.

Firstly, we combined the **Archard wear law** and established the concept of the maximum wear rate, and derived and built a **Wear Degree** model similar to **normalization**. To verify this model, we used **the Gaussian distribution** superposition method to generate virtual step wear depth distribution data, simulating the combined effects of human trampling and natural wear. This enabled us to construct a **Wear Degree** Visualization Model representation.

The formula based on the wear law can quantify the relationship between wear and usage time. Through formula derivation and non-destructive measurement, the usage frequency of the staircase can be obtained. Based on the two-dimensional planar staircase wear heat map in the visualization model, indicators for judging the preference for going up or down the stairs were obtained by considering the different main force application areas when going up and down the stairs. Based on the concept of **Wear Degree**, **FCE** was used, and the usage mode of the staircase by multiple people was determined by the evaluation level of the maximum value in the comprehensive evaluation results. In the formula model, environmental correction factors were added to make the model more consistent with actual usage conditions. In the visualization model, the sensitivity of the model was improved by increasing the wear center and diffusion standard deviation.

Regarding the issue of time, for the uneven wear of each material over time, a three-dimensional image of the **Wear Degree** change of the material over time was established to guide archaeologists in making necessary corrections when using the model to determine the age.

In terms of construction materials, when determining whether the material has been repaired, methods such as 3D scanning and camera depth of field analysis were used to obtain data. The repair was judged by whether the change in **the curvature radius** of the worn area was smooth and whether the inner angles of the staircase corners were consistent with those before repair. When determining the repair material of the staircase, non-destructive testing methods such as ultrasonic and X-ray detection were adopted. For wood, the tree rings and other features could be observed to determine if the replacement material was consistent with the previous one.

Finally, to further determine the usage mode of the staircase, the concepts of wear center density and wear center depth were used to analyze the three-dimensional wear depth heat map, combined with the two-dimensional planar wear heat map, to reflect the usage pattern of the staircase well.

Keywords: Archard wear law; normalization; Wear Degree ; Gaussian Distribution Superposition; the curvature radius; FCE

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1 Introduction

1.1 Problem Background

Stairs, as a basic part of building structures, have long been subjected to constant use and wear. The materials of construction (such as stone or wood) are gradually degraded by long-term exposure to human and environmental factors. This wear not only reflects the frequency of use of the stairs, but also provides valuable information about usage patterns, flow directions, and historical context. For example, uneven wear on stairs, such as a deep depression in the center of a step, can indicate whether people are primarily moving in a particular direction (up or down), or whether the stairs are being used by more than one person at a time. In addition, the intensity and distribution of wear can reveal whether the stairs are used by a large number of people in a short period of time or by a small number of people over a longer period of time. Archaeologists and historians can use this information to infer the history of the building, pedestrian patterns, and possible renovations. However, accurately determining the age of a building or when it was built remains challenging, especially if the site has been occupied for a long period of time prior to the building being built. In addition, revisions, repairs, and the addition of new structural elements over time further complicate the analysis.

To address these challenges, it is necessary to develop a robust model that can analyze stair wear patterns and provide historical insights into their use. The model should take into account material properties, environmental conditions (e.g. weathering, Chemical Corrosion) and human activities (e.g. pedestrian flow patterns). By combining mathematical modeling, numerical simulation and non-destructive measurement techniques, we can reconstruct the historical use of stairs and provide valuable guidance for archaeological and historical research. Not only does this approach help to understand the past, it can also predict future wear and tear, and provide a basis for planning for maintenance or restoration of historic buildings.

1.2 Restatement of the Problem

Considering the background information and restricted conditions identified in the problem statement, we need to solve the following problems:

- **Problem 1**: How often were the stairs used? And how to measure the frequency of stair use?
- **Problem 2**: Was a certain direction of travel favored by the people using the stairs?
- **Problem 3**: How many people use stairs at the same time? The way they use stairs, they tend to climb the stairs side-by-side or did they travel single file ?
- **Problem 4**: Does wear consistent match the historical data and the information available ? What other external potential factors affect stair wear?
- **problem 5**: How to determine the service life of stair well by the degree of wear? How do you think that makes predictions more reliable?

- **Problem 6**: How to determine if the stairs have undergone restoration or renovation work? And its subsequent service life?
- **Problem 7**: Can the source of the material be determined?
- **Problem 8**: Does the intensity of wear reflect the different numbers of people passing through?(the large numbers of people using the stairs over a short time or a small number of people over a longer time)

1.3 Our Works

- We design a Wear Degree Model, which considers that stairs will be difficult to use or the frequency of use will be greatly reduced after a certain degree of wear compared with intact stairs(need renovations), then there is a Maximum Wear Amount, and the mathematical expression model of Wear Degree is deduced from this Wear Amount combined with the Total Wear Amount and Wear Law.
- We generate virtual step wear depth distribution data by numerical simulation, simulate the coupling effect of artificial tread and natural wear by using Gaussian Distribution Superposition Method, Construct Wear Depth Model, and construct visual image of stairs.
- Through **Mathematical Analysis**, **Visualization Analysis and Fuzzy Comprehensive Evaluation method**, we get the preference pattern of stairs in use.(Preference for going up and down stairs, preference for how to use stairs simultaneously)
- Discuss the influence of other factors such as Chemical Corrosion and Wear Fatigue on the above two models. By discussing the factors affecting coefficient Kin Wear Law, the qualitative change index of Wear Coefficient K is obtained, and the deviation between calculated wear information and actual situation is determined.
- Wear degree models are used to establish wear versus time to estimate the age of stairwells and predict future use and repair conditions. If it is judged that the stairs have undergone repair, the wear data should be analyzed for the second time, including Wear Amount before repair and Wear Amount after repair, as well as whether the use of materials before and after repair is consistent. Finally, the service life is comprehensively obtained to predict the future service condition.
- For structural changes after stair wear, use measurement methods to determine the change in the radius of curvature of the worn arc, and determine whether renovation or maintenance is carried out by measuring whether the internal angle of the angle is within the normal deviation range. Combined with the Wear Degree Model and the use of materials, through the subsequent analysis of the whole material[4] after repair or renovation, it is determined whether there is a difference between the repaired or refurbished material and the original material, and whether the repair mutation point has stability, so as to realize the reprediction of subsequent service life.

- By analyzing the Wear Degree Visualization Model and studying its mathematical characteristics, combined with the use pattern of stairs obtained through measurement in a period of time, it is judged whether stairs are used in large numbers of people over a short time or a small number of people over a longer time .

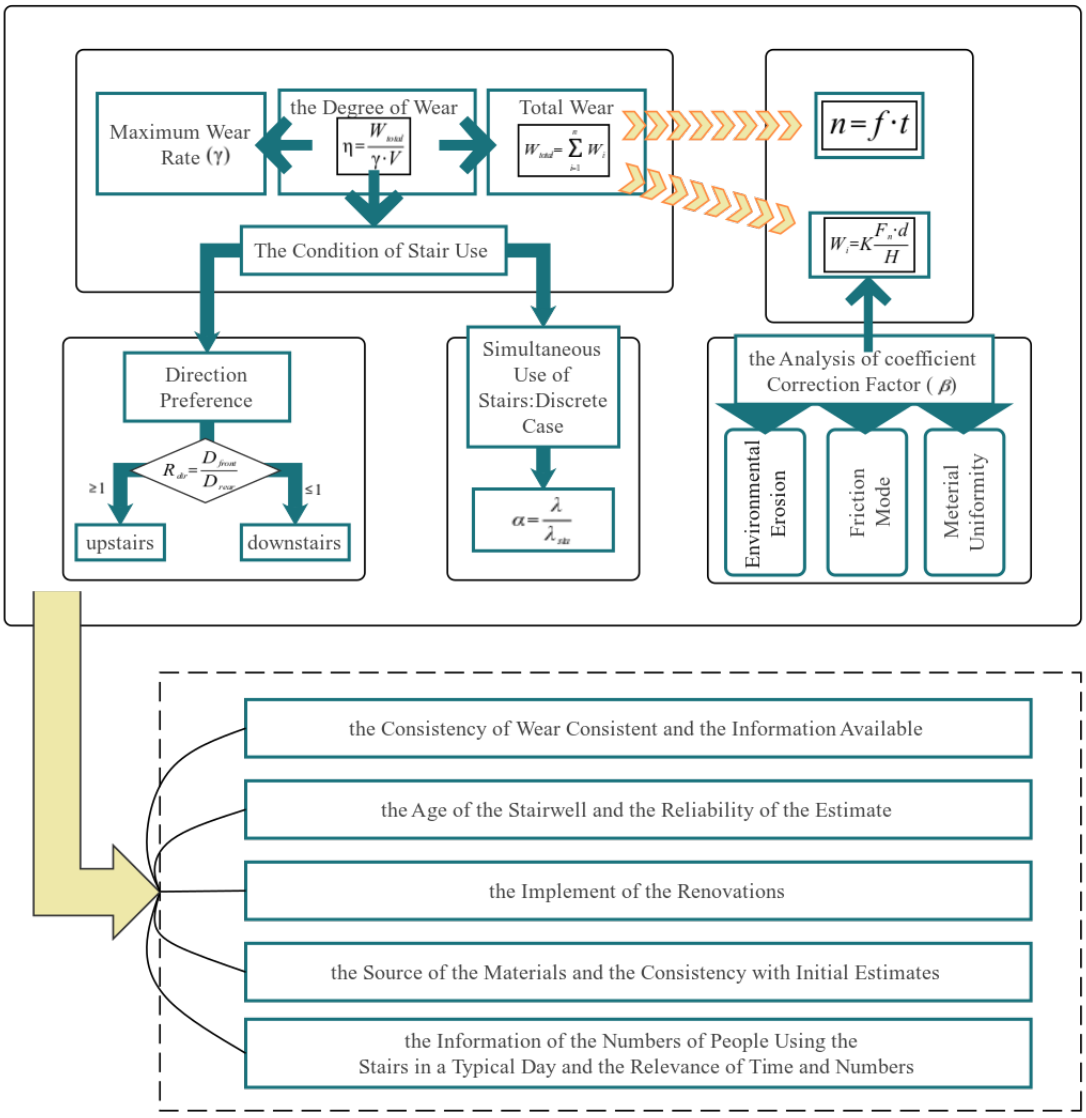


Figure 1: Mind Mapping

2 General Assumptions and Variable Description

2.1 General Assumptions

To simplify the given problems and modify it more appropriate for simulating real-life conditions, we make the following basic hypotheses, each of which is properly justified.

- **Assumption 1**: In establishing the concept of Maximum Wear Rate, it is assumed that stairs will be difficult to use or unusable .
  Reason:Continuing to use it might be dangerous; it must be refurbished.

- **Assumption 2**: In describing the concept of Wear Degree, it is assumed that the stair material (e.g. stone or wood) is homogeneous in the unrepaired area and that the mechanical properties (e.g. hardness, wear resistance) are consistent in all directions.
  Reason: Using the same piece of material to construct stairs is relatively uniform, simplifying the Wear Coefficient calibration process.

- **Assumption 3**: The frequency of stair use (daily average number of passengers) and the direction of passage remain stable on the time scale, and there is no short-term surge in people flow caused by unexpected events (such as war and celebration).
  Reason: Long-term wear analysis usually relies on Averaging Effect and ignores short-term fluctuations.

- **Assumption 4**: Wear is mainly caused by human activities (treading). In practice, weathering and treading may be non-linearly coupled, but linear separation is convenient for parameter calibration. The effects of natural weathering (such as rain erosion and temperature change) can be linearly superimposed by environmental correction factors.
  Reason: simplify the detection logic of repair traces (such as judging repair time only by mutation point analysis). If there is a significant difference in repair materials (e.g. cement repair stone steps), a material comparison module needs to be introduced.

- **Assumption 5**: Forefoot force[2] is fixed at 60% for upward movement and heel force at 70% for downward movement. When many people walk side by side, the pressure is evenly distributed and there is no lateral extrusion force.
  Reason: Set typical values based on biomechanical studies, ignoring individual differences.

- **Assumption 6**: Measuring instrument tool errors meet factory standards, ignoring operational errors or extreme environmental disturbances. If impact loads or corrosive media (e.g. seawater) are present, the wear mechanisms need to be extended, e.g. correction factors.
  Reason:Simplifies model complexity and facilitates analytical solution derivation.

2.2 Variable Description

We list the symbols and notations used in this paper in Table 1.

Table 1: Symbols and Definitions

Symbols	Definitions
γ	Maximum Wear Rate
W_{total}	Total Wear Amount
η	Wear Degree
V	step volume
W_i	wear per time
f	Wear Frequency
t	time
n	wear times
H	material hardness
F_n	wear surface vertical force
d	Distance per slide
K	Wear Coefficient
λ	standard discrete distance
α	dispersion
β	correction factor
R_{dir}	directivity index
B	comprehensive evaluation result
r	radius of curvature

3 Model Establishment and Analysis

3.1 Model Establishment

3.1.1 Formula Construction of Wear Degree Model

For the wear accumulation process of the step, there will be a Wear Amount Wi for each wear. After the step is worn to a certain ratio, the step will not be used. At this time, there will be a maximum Wear Amount.As shown in Figure 2.The ratio of this Wear Amount to the total volume of the step is called Maximum Wear Rate γ.

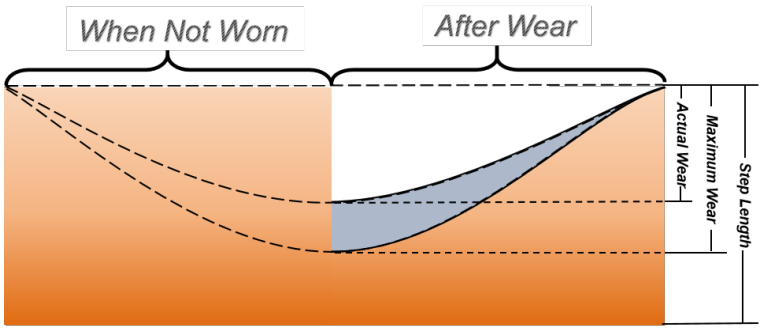


Figure 2: the Degree of Wear

$$\gamma = \frac{W_{total}}{W_{totalmax}} \tag{1}$$

For single Wear Amount, **Archard wear law**[1] is introduced.

$$W_i = K\frac{F_n \cdot d}{H} \tag{2}$$

We set the Wear Frequency f in a period of time t, and combined with the Wear Law, we can get the Total Wear Amount W_total in a period of time. According to Total Wear Amount, Maximum Wear Rate and the volume of the step itself, a formula 3 for evaluating Wear Degree is obtained.

$$\eta = \frac{W_{total}}{\gamma \cdot V} \tag{3}$$

The derivation process of the whole formula 3 is shown in Figure 3.

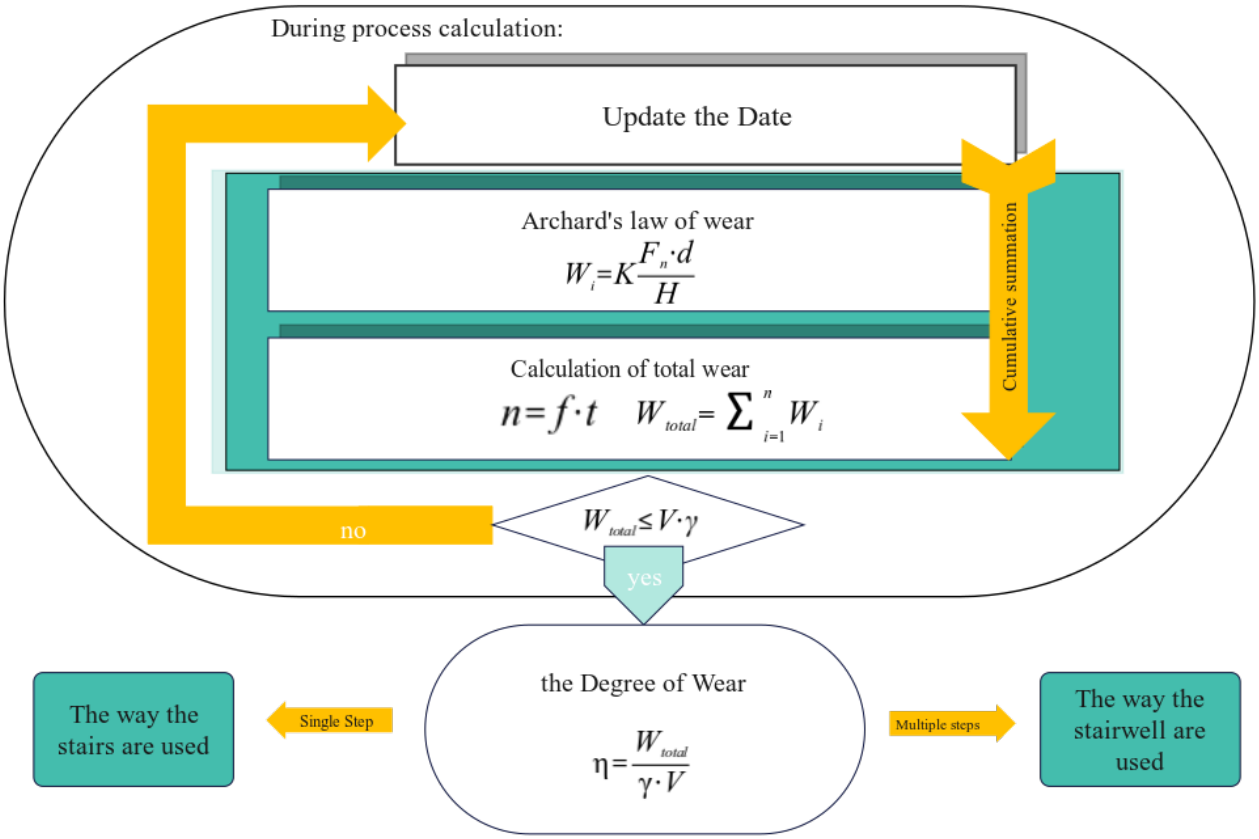


Figure 3: the derivation of wear degree

3.1.2 Simulation Construction of Visual Model of Wear Degree of Single Step

In the modeling process, to validate the reliability of **Wear Degree Model**, we generated virtual step wear depth distribution data through numerical simulations.

1. Geometric Parameter Definitions
Based on historical documentation, we selected a physical scale of **Length = 350 mm**, **Width = 150 mm**, and **Height = 170 mm** to simulate a typical stone step. Drawing from extreme wear cases observed in historical structures, we set the maximum allowable wear depth at **40 mm** as an upper limit. To balance computational efficiency and detail preservation, we assumed **100 wear centers** to account for localized wear accumulation caused by repeated pedestrian trampling on fixed areas.
2. Wear Depth Model Construction
We adopted the **Gaussian Distribution Superposition Method** to simulate the coupled effects of human trampling and natural abrasion. The model integrates:

- **Uniformly distributed micro-noise** (simulating natural weathering and minor friction).
- **100 local concave wear centers** generated by Gaussian functions, superimposed onto the noise layer. Each center was constrained to a maximum depth of **40 mm** to avoid physically unrealistic results.

3. Gaussian Distribution Parameter Settings
Key parameters for each Gaussian distribution include:

- **Center Position:**

  70%**of wear centers** were clustered in the middle region near the step edge, reflecting pedestrian walking patterns along the central axis.

  30% **were randomly distributed**, simulating occasional path deviations or multi-person parallel trampling.

- **Depth:**

  Based on long-term wear records of stone steps, significant concave depths typically range from $15 \sim 40mm$.

- **Lateral Diffusion Standard Deviation:**

  $\sigma_x, \sigma_y \sim U(5, 15)$ mm, referencing biomechanical studies indicating that adult single-step pressure diffusion spans approximately $10 \sim 30mm$.

4. Data Generation and Visualization
The final output comprises a dataset of **(x, y, depth)** covering the entire step surface (grid resolution: **0.2 mm**). Representative data as shown in Table 2. For intuitive analysis, we generated:

Table 2: Partial Simulation Data of Step Surface Wear

X (mm)	0.40	2.60	107.8	115.40	170.2	159.6	221.6	225.40
Y (mm)	0.00	0.00	7.8	11.80	13.40	35.8	44.20	44.20
Depth (mm)	22.59	22.18	6.94	31.68	40.00	40.00	40.00	31.35

- **Planar Heatmaps (Figure 4):** Visualizing spatial wear distribution.
- **3D Histograms (Figure 5):** Displaying depth variations across the step.

5. Data Validation
As shown in **Figure 4**, the central region exhibits significantly greater depths than the edges, aligning with the "central concavity" phenomenon observed in ancient architectural staircases. **Figure 5** demonstrates a right-skewed depth distribution, consistent with statistical features of real-world wear data:

- **Shallow regions (**$010mm$**)**: 52.04%, corresponding to natural weathering and low-frequency trampling.
- **Deep regions (**$\geq 30mm$**)**: 16.53%, reflecting concentrated wear from high-frequency pathways.

6. Theoretical and Practical Implications
Planar Heatmap (Figure 4):

- **Theoretical Significance:** Reveals spatial patterns of pedestrian pathways, supporting directional inference (e.g., central concavity indicates unidirectional movement). Quantifies wear concentration to prioritize repair efforts.
- **Application:** Used in expert presentations to highlight high-wear zones and guide preservation planning.

3D Histogram (Figure 5):

- **Theoretical Significance:** Validates the Gaussian diffusion hypothesis (histogram profiles match theoretical curves). Identifies abnormal depressions (e.g., isolated deep pits), signaling potential localized damage or repair traces.
- **Application**: Academic comparisons between simulated and measured wear morphologies.

This methodology ensures robust validation of stair wear models while providing actionable insights for heritage conservation and structural maintenance.

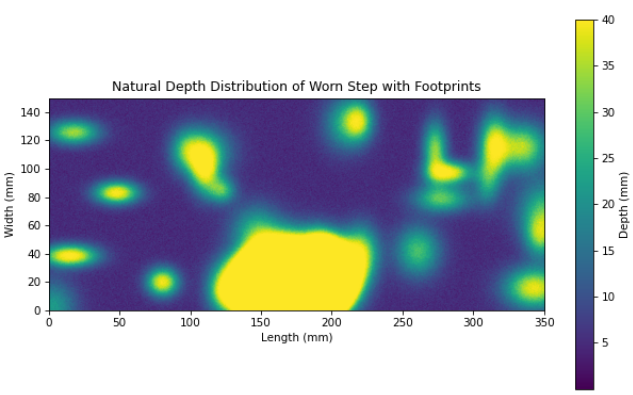


Figure 4: Planar Heatmap

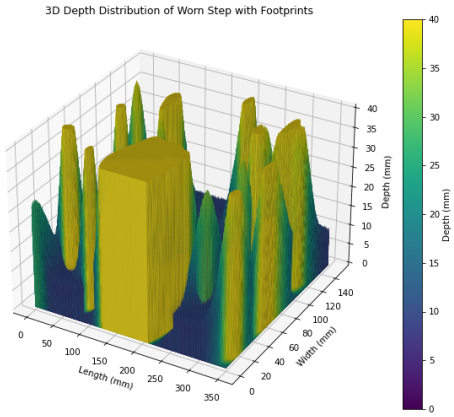


Figure 5: 3D Histogram

3.2 Preliminary use of the model

3.2.1 Frequency of Stairs

We can measure the wear volume of each step through 3D scanning equipment tools or cameras combined with magnification and depth of field information. Combined with the materials used to manufacture stairs, the length of use of stairs and Wear Amount, when each Wear Amount $W_i$is uniform, we can objectively calculate the wear of each stair. The Formula 4 is derived as follows.

$$f = \frac{W_{total} \cdot H}{K \cdot F_n \cdot d} \tag{4}$$

Solve for frequency of use f(person-times/second).
After many repairs, according to whether the materials used are the same or not, they can be calculated in stages and then summed. Formula 5 is as follows

$$f = \sum_{1}^{m} \frac{W_m \cdot H}{K_m \cdot F_n \cdot d} \tag{5}$$

3.2.2 Direction Preference

Divide the single-step Wear Degree Visualization Model into two parts, namely, the upper floor key area and the lower floor key area, according to the two-dimensional plane thermal diagram(Figure 2) generated in 3.1.2. Calculate the Wear Amount of the upper floor key area and the lower floor key area respectively according to the different analysis

of the upper and lower floor key area, and calculate Wear Rate. Design the directivity index according to the ratio of Wear Rate of the upper and lower floors.

$$R_{dir} = \frac{\eta_{front}}{\eta_{rear}} \tag{6}$$

$R_{dir} > 1$,upstairs dominance

$R_{dir} < 1$,downstairs dominant

Through the verification method of directional index, the conclusion of which direction people prefer to walk is obtained. By the difference between the up-and-down Wear Rate and the complete Wear Rate, it is possible to simply predict the safe passenger flow in the future up-and-down.

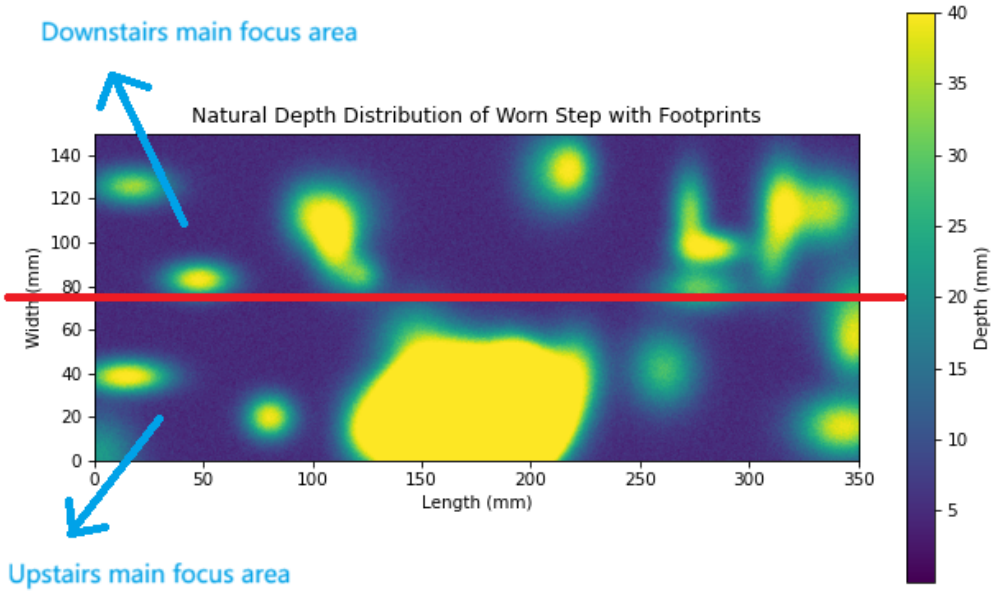


Figure 6: Upstairs and downstairs mainly focus on the division of areas

3.2.3 Simultaneous Use of Stairs

In order to judge the historical traffic pattern use by the wear characteristics of stairs, we use the Fuzzy Comprehensive Evaluation Method.

First determine the set of factors $U = \{U_1, U_2, U_3, U_4\}$.

- U_1:Lateral wear uniformity (wear difference between left and right sides of step)
- U_2:Longitudinal wear concentration (wear distribution of leading edge and trailing edge)
- U_3:Wear center density (ratio of wear pit area greater than a certain Wear Degree to upper surface area of step)
- U_4:Wear center depth (normalized to Wear Degree greater than a certain threshold)

Factor set determines evaluation set $V = \{V_1, V_2, V_3\}$, determines weight set, and assigns each factor weight through expert scoring or historical use statistics.

- V_1: side-by-side walking mainly
- V_2:single-person walking mainly
- V_3:mixed model

weight vector $W = \{W_1, W_2, W_3, W_4\}$

- W_1:transverse uniformity
- W_2:vertical concentration
- W_3:footprint density
- W_4:wear center depth

Construct membership function by constructing fuzzy membership function for each factor, actual measurement values are mapped to evaluation grades.

$$\begin{Bmatrix} U_1 & U_2 & U_3 & U_4 \end{Bmatrix} = \begin{cases} \mu_{V_i}(U_1) = \begin{cases} 0 & ,U_1 \le 0.3 \\ \frac{U_1 - 0.3}{0.7 - 0.3} & ,0.3 < U_1 < 0.7 \\ 1 & ,U_1 \ge 0.7 \end{cases} \\ \mu_{V_i}(U_2) = \begin{cases} 1 & ,U_2 \le 0.4 \\ \frac{0.8 - U_2}{0.8 - 0.4} & ,0.4 < U_2 < 0.8 \\ 0 & ,U_2 \ge 0.8 \end{cases} \\ \mu_{V_i}(U_3) = \begin{cases} 0 & ,U_3 \le 0.3 \\ \frac{U_3 - 0.3}{0.7 - 0.3} & ,0.3 < U_3 < 0.7 \\ 1 & ,U_3 \ge 0.7 \end{cases} \\ \mu_{V_i}(U_4) = \begin{cases} 1 & ,U_4 \le 0.2 \\ \frac{1.0 - U_4}{1.0 - 0.2} & ,0.2 < U_4 < 1.0 \\ 0 & ,U_4 \ge 1.0 \end{cases} \end{cases}$$

For each factor U_i,the membership degree to each evaluation grade$V_j$is calculated to form a matrix.

$$R = \begin{bmatrix} \mu_{V_1}(U_1) & \mu_{V_2}(U_1) & \mu_{V_3}(U_1) \\ \mu_{V_1}(U_2) & \mu_{V_2}(U_2) & \mu_{V_3}(U_2) \\ \mu_{V_1}(U_3) & \mu_{V_2}(U_3) & \mu_{V_3}(U_3) \\ \mu_{V_1}(U_4) & \mu_{V_2}(U_4) & \mu_{V_3}(U_4) \end{bmatrix}$$

A comprehensive evaluation result B is obtained by synthesizing weight vector W and fuzzy matrix R.

$$B = W \circ R = \begin{bmatrix} b_1 & b_2 & b_3 \end{bmatrix} \tag{7}$$

where $\circ$ is a fuzzy composition operator.

$$B_j = \sum_{i=1}^{m} \left[W_i \cdot \mu_{V_j}(U_i) \right] \tag{8}$$

Result judgment: Take the evaluation grade corresponding to the maximum value in B as the final conclusion:

- $b_1 = max(B) \rightarrow$side-by-side walking mainly
- $b_2 = max(B) \rightarrow$single-person walking mainly
- $b_3 = max(B) \rightarrow$mixed mode

3.3 Stability and Sensitivity Analysis

Through the detailed analysis in Sections 3.2.2 and 3.2.3, we can preliminarily determine the usage patterns of the stairs. These patterns are crucial for understanding the daily activities related to the stairs. However, it should be noted that both the **Wear Degree Model** and the **Visual Model of Wear Degree of Single Step** are still subject to external and internal influencing factors.

Specifically, external factors such as environmental conditions and foot traffic volume, and internal factors like the material composition and manufacturing process, all play significant roles in affecting these models.

Moreover, in the Formula Model established in Section 3.1.1(Formula 2), the wear coefficient may be affected by **Fatigue Wear** or changes in the surface properties of the material (e.g., Chemical Corrosion). For instance, during the initial wear stage, the actual contact area between the step edge and the shoe sole is small, and the microscopic surface roughness leads to a faster wear rate. As wear progresses, the contact area increases, and the microscopic surface gradually smooths out, transitioning into the normal wear

stage, where the wear rate slows down. When the wear amount reaches a certain level, the material's **Wear Resistance Coefficient** may decrease, causing the wear coefficient to accelerate and enter the severe wear stage. Consequently, the wear rate exhibits a characteristic of first decreasing and then increasing, as shown in the Figure 7 below:

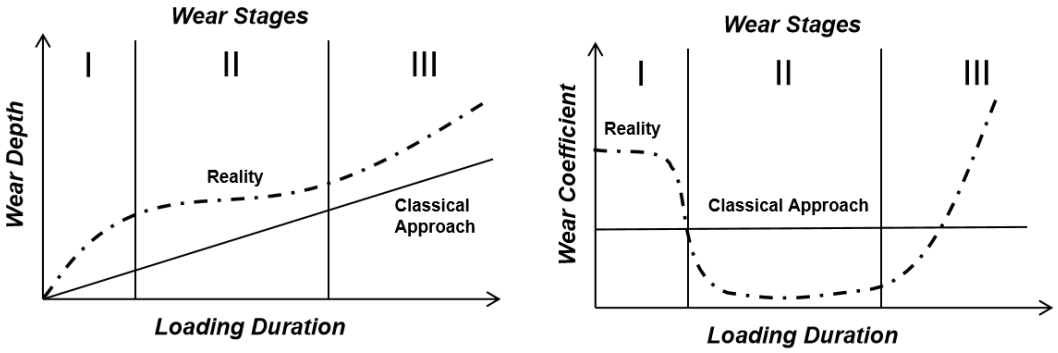


Figure 7: loading duration

Due to the progression of the friction stage and the inherent inhomogeneity of the material, the combination of Sliding Friction and **Static Friction** occurs, as shown in the Figure 8 below:

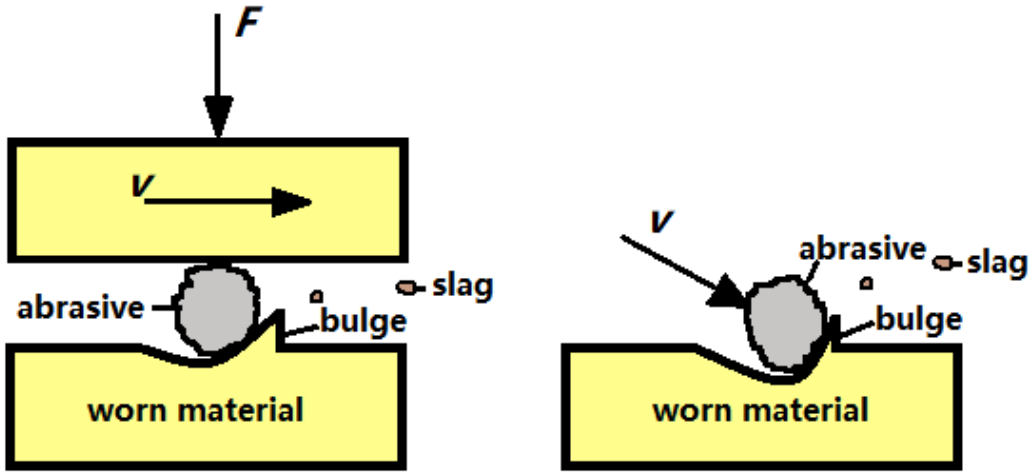


Figure 8: Material Wear Force Analysis Diagram

The corresponding changes in wear depth across different stages are shown in the following Figure 9:

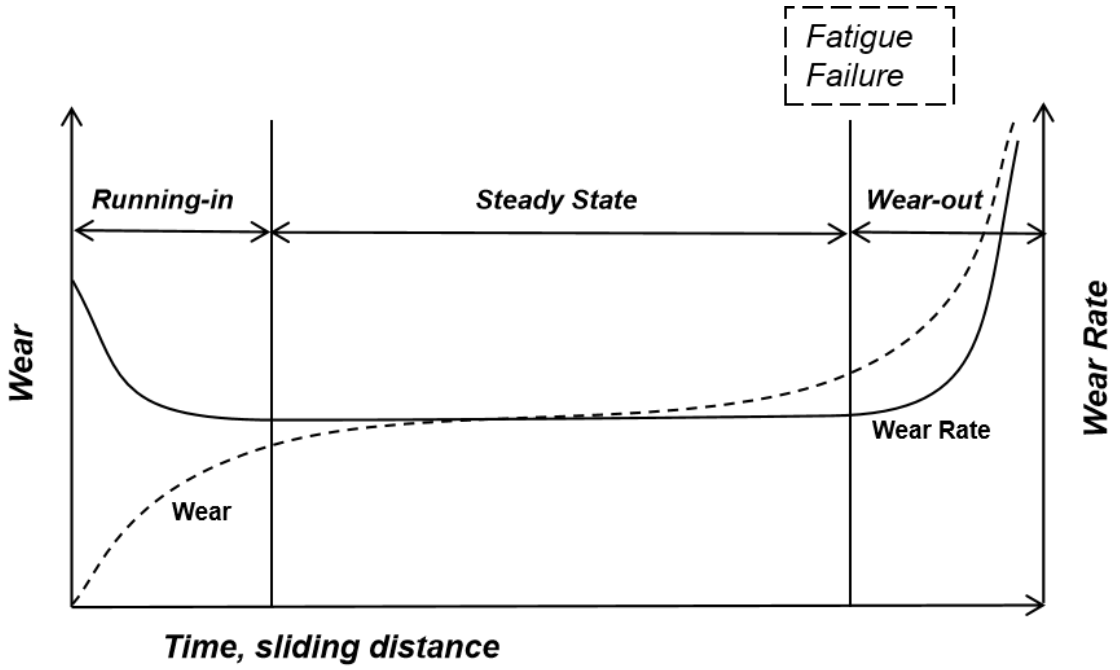


Figure 9

Additionally, the **Daily Usage Frequency** (average daily foot traffic) and **Direction of Movement** may not remain stable over time. For example, sudden events (e.g., wars, celebrations) can lead to short-term surges in pedestrian flow.
The above issues can be addressed by incorporating environmental correction factors(β) for adjustment.

$$W_i = K\frac{F_n \cdot d}{H}\beta \tag{9}$$

During the construction of the Visual Model of Wear Degree of Single Step, a Parameter Sensitivity Analysis was conducted:

- **Number of Wear Centers:** Increasing the number of wear centers from 50 to 150 resulted in the average depth of the central region increasing from **28.5 mm** to **34.1 mm**, consistent with linear cumulative expectations.
- **Diffusion Standard Deviation:** If $\sigma_x, \sigma_y$are reduced to 2–5 mm, the depression contours become sharper. Further adjustments to the model parameters are required to align with real-world data from "heavy-step gait" scenarios (e.g., armory stairs).

4 model extension

4.1 the Solution of Temporal Problem

4.1.1 Wear Consistent with Available Information

Wear Consistent, which was analyzed for model sensitivity in section 3.3, requires environmental correction factors to be consistent with available information collected by archaeologists or historians.

4.1.2 Age Judgment and Reliability of Stairwell

In section 3.2.1, we have derived the formula for stair frequency using formula model, such as Formula 4.We can change the formula a little bit to get the age of the stairwell, as shown in the Figure 10.

$$t = \frac{W_{total}}{W_i \cdot f} \tag{10}$$

Using Wear Amount to calculate Wear Degree, we can get the two-dimensional graph of Wear Degree and service life of this type of stairs, as shown in the following Figure 10.If archaeologists analyze different stair building materials and obtain the Wear Degree and Duration of each material, they can construct a three-dimensional column chart of Wear Degree of each material and the duration of stair use,as shown in the Figure 11.

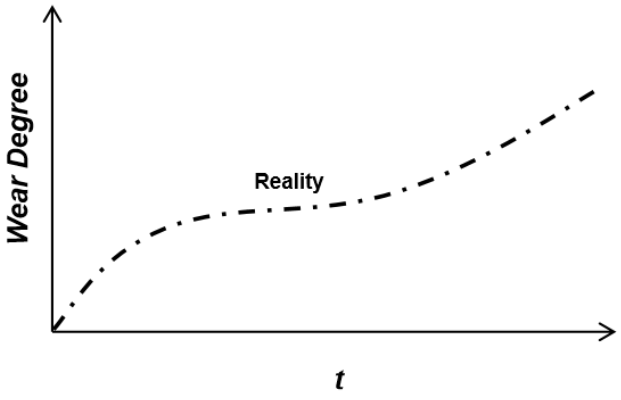


Figure 10: 2D single material

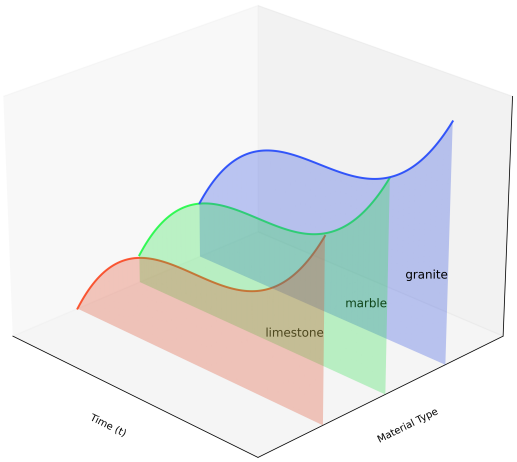


Figure 11: 3D Three kinds of materials

Then calculate the comprehensive service life after repair with various materials, and predict the sustainability of subsequent repair with various materials.

4.2 Renovation and material problem solving

4.2.1 Judge whether to renovate

To judge whether the stairs are renovated, you can judge the repair status by the method of angle radian, using three-dimensional laser scanner, portable radian measuring instrument, image processing software, etc.

If there is an unrepaired stair run, the unrepaired stair run can be used as a benchmark, and the first repair area can be judged by the method after radian feature extraction.

- Curvature analysis:calculating radii of curvature at corners using 3D scan data R:

$$r = \frac{\left(1 + (f')^2\right)^{3/2}}{|f''|} \tag{11}$$

  Judgment criteria: normal wear area curvature radius changes gently in line with the natural wear law, repair area curvature radius mutation or inconsistent with the surrounding area.

- Pieces angle contrast: You can also measure the edges and corners of the interior angle (the angle between the leading edge of the step and the side),the repair may lead to angle deviation.

4.2.2 Refurbishment of materials

Using the three dimensional graph table (Figure 11)a constructed in section 4.1.2, we can judge that the similarity between the Wear Degree of the refurbished material and the raw material within a certain period of time can be analyzed. The similarity error can be judged to be the same material within an acceptable range.

Non-destructive testing methods can be used where conditions permit, such as ultrasonic testing X-ray testing, to detect the structural consistency of the material; if the raw material is wood, it can also be observed to determine whether the wear is consistent with the age and type of wood originally assumed to be used.

4.3 Deep promotion analysis of usage patterns

In section 3.2.3, the concept of wear center density and wear center depth is introduced. Combined with the two-dimensional plane wear thermal diagram(Figure 4), we set a lower wear depth threshold by analyzing the three-dimensional cylindrical thermal diagram of wear depth(Figure 5). Under this threshold, the wear distribution and depth are analyzed.

The wear center density is small, the wear center depth is shallow, especially the step edge angle part, if the wear is more uniform and shallow, it can be understood as a small number of people for a long time; the wear center density is large, the wear center depth

is deep, and the wear is serious and concentrated in some areas. The stairwell design can also provide some information for this purpose. The stairwell is narrow and the passage is not smooth, which is more suitable for a small number of people to use for a long time. The stairwell is wide and the passage is smooth, which may be easier to accommodate a large number of people to use in a short time. We can make some non-destructive measurements on the stairwell to record the flow of people, the distribution of people and the time pattern over a period of time. These data can directly reflect the use pattern of the stairwell.

5 Conclusion and Future Work

We establish a Wear Degree Model, analyze the use of steps through formulaic and visual angles, predict the use and repair experience of steps, predict the future use and possible renovation of steps, and use the concept of Wear Degree in the model, which is similar to normalization, which can improve the convergence speed of the model in calculation, improve the accuracy of the model, and enhance the stability of the model.It can also more intuitively reflect the used condition of the stair and predict the future repair situation. The model uses many linear assumptions, but considering that the internal and external factors of the material are nonlinear transformation, the Wear Coefficient K value presents a nonlinear transformation trend and cannot be constant, so it is necessary to add environmental correction factors to perfect the model to improve the sensitivity of the model. There are still subjective factors in defining standards and thresholds for some data, which may affect the objectivity and accuracy of evaluation results.Fuzzy Comprehensive Evaluation Method is used in the analysis process. The calculation process of this method is relatively complex, which will increase the complexity of solving some problems by the model.

References

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[4] Lourenço, P. B. (2004). Analysis and restoration of ancient masonry structures: guidelines and examples. In INNOVATIVE MATERIALS AND TECHNOLOGIES FOR CONSTRUCTION AND RESTORATION (Vol. 4, pp.). University of Lecce. Department of Innovation Engineering. This paper provides methods for material characterization and analysis, which can be used to determine the source of the stair material.

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Appendices

Python code used in this article is shown below:

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D
from matplotlib import cm
from matplotlib.colors import Normalize
import pandas as pd

# Define the dimensions of the step
length = 350  # Length (mm)
width = 150  # Width (mm)
max_depth = 40  # Maximum depth (mm)
num_footprints = 100  # Number of footprints

# Create a grid
x = np.arange(0, length, 0.2)  # Length is 350mm, step size is 0.2mm
y = np.arange(0, width, 0.2)  # Width is 150mm, step size is 0.2mm
X, Y = np.meshgrid(x, y)  # Create a length - width grid

# Define the depth calculation function
def calculate_depth(X, Y, num_footprints):
depths = np.random.rand(*X.shape) * (max_depth / 4)  # Basic noise
center_x = length / 2
center_y = width / 8

for _ in range(num_footprints):
if np.random.rand() < 0.7:
center_x_pos = np.random.uniform(center_x - 30, center_x + 30)
center_y_pos = np.random.uniform(center_y - 15, center_y + 15)
else:
center_x_pos = np.random.uniform(0, length)
center_y_pos = np.random.uniform(0, width)

max_depth_for_footprint = np.random.uniform(15, max_depth)
sigma_x = np.random.uniform(5, 15)
sigma_y = np.random.uniform(5, 15)

footprint_depth = max_depth_for_footprint * np.exp(-(((X - center_x_pos) ** 2) / (2 * s
((Y - center_y_pos) ** 2) / (2 * sigma_y ** 2)))
depths += footprint_depth

depths = np.clip(depths, 0, max_depth)
return depths

# Calculate the depth data
depths = calculate_depth(X, Y, num_footprints)

data = {
    'X (mm)': X.flatten(),
```

```
    'Y (mm)': Y.flatten(),
    'Depth (mm)': depths.flatten()
}

# Create a DataFrame
df = pd.DataFrame(data)

# Add a description line
description = "This table shows the depth values corresponding to each (x, y) coordinat

# Save to a CSV file
with open('depth_data_with_description.csv', 'w', encoding='utf-8') as f:
# Write the description
f.write(description + '\n')  # Add the description and a new line
# Write the DataFrame's header and data
df.to_csv(f, index=False)  # Add the data to the file

# Plot a 2D heatmap
plt.figure(figsize=(10, 6), dpi=75)
heatmap = plt.imshow(depths, extent=[0, length, 0, width], origin='lower', cmap='viridi
plt.colorbar(heatmap, label='Depth (mm)')
plt.title('Natural Depth Distribution of Worn Step with Footprints')
plt.xlabel('Length (mm)')
plt.ylabel('Width (mm)')
plt.grid(False)
plt.xlim(0, length)
plt.ylim(0, width)
plt.show()

# Plot a 3D bar chart
fig = plt.figure(figsize=(12, 8), dpi=75)
ax = fig.add_subplot(111, projection='3d')

# Create bar chart data
x_pos = X.flatten()
y_pos = Y.flatten()
z_pos = np.zeros_like(depths.flatten())

# Width and depth of each bar
dx = dy = 0.2  # Adjust the width according to the grid
dz = depths.flatten()

# Generate colors for each bar, using the same color map as the heatmap
norm = Normalize(vmin=0, vmax=max_depth)
colors = cm.viridis(norm(dz))

# Plot the bar chart
ax.bar3d(x_pos, y_pos, z_pos, dx, dy, dz, color=colors, zsort='average')

# Set axis labels
ax.set_xlabel('Length (mm)')
ax.set_ylabel('Width (mm)')
ax.set_zlabel('Depth (mm)')
ax.set_title('3D Depth Distribution of Worn Step with Footprints')
```

```
    # Add a color bar, consistent with the heatmap
    mappable = cm.ScalarMappable(norm=norm, cmap='viridis')
    mappable.set_array(dz)
    cbar = fig.colorbar(mappable, ax=ax, label='Depth (mm)')

    # Display the figure
    plt.show()
```

AI Usage Report
Team Control Number: 2515586
Problem Chosen: 2025 MCM Problem A

1. Specific Applications of AI Tools

1. **ChatGPT-3.5 for Programming and Data Processing Assistance**
    - Used to generate initial code for Python plotting and data distribution functions (e.g., Gaussian distribution). The team debugged and optimized the generated code to ensure it met practical requirements and verified the accuracy of the results.
2. **DeepSeek for Problem Understanding and Literature Retrieval**
    - Assisted in understanding the MCM problem, helping the team quickly grasp the core points.
    - Helped locate key references and explained specialized terms and formulas. For example, the explanation of Archard's wear law:
        - **Reference**:Archard, W. (1953). Research in the Journal of Research in Friction Materials. American Materials Scientists.This paper presents the Archard wear law, which is used in your wear degree model.
          Mathematical Society and Addison-Wesley
          Publishing Company , 1984-1986.
        - **Explanation**: Archard wear law describes the relationship between material wear volume, contact pressure, sliding distance, and material hardness, expressed as $W = K\frac{F_n \cdot d}{H}$, where W is wear volume, K is the wear coefficient, F_n is the normal force, d is the sliding distance, and $H$is the material hardness.
3. **Kimi for Data Compilation and Analysis**
    - Used Kimi to compile and analyze all collected references, data, and team-summarized methodologies, helping the team sort out research ideas.
    - Validated the logic of the team's wear degree model to ensure its rationality and scientific rigor.
4. **DeepSeek for Online Tool Price Comparison**
    - Used DeepSeek to compare prices of relevant tools, helping the team select cost-effective equipment. For example, querying prices of portable 3D scanners and comparing the accuracy and cost of different models.
5. **ChatGPT-4.0 for Language Polishing**
    - Used ChatGPT-4.0 to refine certain sentences, improving the language quality of the paper. For example:
        - Original: "We use some factors to decide how people walk."
        - Polished: "The fuzzy comprehensive evaluation system incorporates four critical factors (U1-U4) to characterize pedestrian flow patterns."

2. Team's Main Responsibilities

1. **Core Model Development**
    - The team independently derived the theoretical framework and formulas for the wear degree model.
2. **Data Collection and Processing**
    - The team collected and processed actual data, verifying and correcting preliminary data generated by AI.
3. **Model Validation and Optimization**
    - The team conducted multiple validations and optimizations to ensure the model's effectiveness and stability in practical applications.
4. **Paper Writing and Chart Creation**
    - The team was responsible for the overall writing of the paper and the creation of charts, ensuring logical clarity, structural coherence, and adherence to academic standards.

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3. Usage Boundaries

1. **Core Work Independently Completed**
    - Theoretical derivation of the wear degree model
    - Determination of the directional index (R_{dir}) threshold
    - Development of on-site measurement plans
2. **AI Output Verification Mechanism**
    - All reference links were manually verified for accessibility.
    - Generated code was tested line by line.
    - Graphical data were cross-verified using Matlab.